

CS418 HW11

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8.14

Let S be a set of n points.

Point line duality where point $p = (a, b)$ is mapped to the line $p^* : y = ax - b$

Points that are collinear in the primal plane correspond to lines that are concurrent in the dual plane.

Arrangement of n lines can therefore be constructed in $O(n^2)$ lines.

If k primal points lie on a line, then k corresponding dual lines all pass through the single dual point ℓ^* . If k dual lines meet at a vertex v , then the k corresponding primal points all lie on the line v^* . So the line containing the maximum number of points in S is the dual of arrangement vertex with max number of incident lines.

Algorithm:

1. Dualize every point $p \in S$ into the line p^*
2. Construct arrangement $A(S^*)$ of the n dual lines
3. For all vertexes v of the arrangement, count how many dual lines intersect with v
4. Let v^* be the dual of the vertex that has the largest count from previous step
5. Output the primal line v^*

Runtime:

- Dualizing points takes $O(n)$
- Constructing line arrangement takes $O(n^2)$
- Arrangement has $O(n^2)$ vertices, so scanning them and finding max concurrency also takes $O(n^2)$
- Total runtime is therefore $O(n^2)$

8.16A

Nonvertical line $\ell : y = mx + b$

Point line duality where

$$p = (a, c) \leftrightarrow p^* : y = ax - c$$

$$\ell : y = mx + b \leftrightarrow \ell^* = (m, -b)$$

Point p is above ℓ iff ℓ^* is above p^*

For segment $s_i = p_i q_i$, the duals p_i^* and q_i^* are two lines

Primal line ℓ intersects s_i iff endpoints p_i, q_i are on opposite sides of ℓ or one lies on ℓ

Through order preservation in dual plane, this is equivalent to ℓ^* lying between the two dual lines p_i^* and q_i^*

Each segment becomes closed strip in the dual plane. Stabber exists iff there is a point in the dual plane contained in all n strips.

Algorithm:

1. Handle the vertical stabbers separately
2. Vertical line $x = c$ intersects the segment s_i iff c lies in the x interval of s_i
3. So vertical stabber exists iff all x intervals have common intersection $\rightarrow O(n)$
4. Dualize all of the segment endpoints, which gives $2n$ dual lines
5. Construct the arrangement of the $2n$ lines
6. Done in $O(n^2)$ time and arrangement has $O(n^2)$ complexity
7. For each segment s_i , define the dual strip D_i as the region between its two endpoint dual lines
8. Traverse the cells of the arrangement, and for each cell keep the number of strips that contain it.
9. Start with one cell and compute count in $O(n)$
10. When moving across an arrangement edge, only one dual line is crossed.
11. Status of corresponding segment changes, so the count updates in $O(1)$ time
12. If some cell is contained in all n strips, output YES. Otherwise, output NO.

Runtime:

- Vertical stabber check takes $O(n)$
- Arrangement construction on $2n$ lines takes $O(n^2)$
- Arrangement traversal and count updates take $O(n^2)$

11.7

Let q be the query point. Boundary of 3D polytope consists of polygonal facets, and surface is viewed as planar graph. With n vertices it only has $O(n)$ edges and facets overall.

Algorithm:

1. Check if q lies on the boundary. For each facet F , test if q lies in the plane of F and inside the polygon F . If yes, then report inside/on the boundary.
2. Otherwise, shoot a ray p from q in a direction that is not parallel to any facet plane and does not pass through any edge or vertex
3. For every facet F , intersect p with the plane of F . If the intersection point lies inside the polygon F , then count it
4. If the number of intersections is odd, then return inside. If it is even, then return outside

Polytope is simple, and its boundary is a closed surface similar to a sphere. So a generic ray that goes across the boundary every time it enters or leaves the polytope. If it starts outside, then the times it crosses is in pairs so it will be even. If it starts inside, it will also be in pairs besides the first time, which makes it odd.

Runtime

Facet F has k_f vertices. Testing if the ray hits F takes $O(k_f)$ time.

Over all facets, $\sum_F k_F = 2E = O(n)$ because the boundary is a planar graph with $O(n)$ edges/faces.

Therefore the total running time is $O(n)$

11.10A

Non vertical planes are $h : z = ax + by - c$

Define h by dual point $\rightarrow h^* := (a, b, c)$

If $h^* = (a, b, c)$ then $(h^*)^* : z = ax + by - c = h$

Therefore $(h^*)^* = h$ and it is 3D analogue of 2d point line duality where dual of the dual returns original object.

Definitions:

Let p be finite set of points in R^3 .

Upper convex hull $UH(P)$ is part of the boundary of $CH(P)$ that is visible from $+\infty$ in the z direction

It is also the set of facets f of $CH(P)$ where the supporting plane of f has all points of P on or below it $\rightarrow$ upper hull is the part supported by lines lying above all points.

Let H be the set of non-vertical planes.

The lower envelope $LE(H)$ is the pointwise minimum of the planes.

$$LE(H) = \{(x, y, z) : z = \min_{h \in H} h(x, y)\}$$

It is also the bottom surface of arrangement of planes in H .

Property: p lies on h iff h^* lies on p^*

Let

$$p = (p_x, p_y, p_z), p^* : z = p_x x + p_y y - p_z$$

$$h : z = ax + by - c$$

$$h^* = (a, b, c)$$

Assume that $p \in h$ so that $p_z = ap_x + bp_y - c$

This is then rearranged to $x = ap_x + bp_y - p_z$

Now evaluate plane p^* at the point (a, b) :

$$z = p_x a + p_y b - p_z = c$$

So point $(a, b, c) = h^*$ lies on p^*

Then, $c = p_x a + p_y b - p_z$

$$\rightarrow p_z = ap_x + bp_y - c$$

This is then the equation that says $p = (p_x, p_y, p_z)$ that lies on the plane $h : z = ax + by - c$

Therefore, $p \in h \leftrightarrow h^* \in p^*$

11.10B

Property: p lies above h if and only if h^* lies above p^* .

Let $p = (p_x, p_y, p_z)$ and $p^* : z = p_x x + p_y y - p_z$

Non vertical plane is $h : z = ax + by - c$

So the dual point is $h^* = (a, b, c)$

Need to show that p lies above $h \leftrightarrow h^*$ lies above p^*

Proof:

In the primal space, p lies above h when

$$p_z > ap_x + bp_y - c$$

$$\rightarrow c > ap_x + bp_y - p_z$$

Then inspecting the dual plane p^* :

At point (a, b) , the z value is $p^*(a, b) = p_x a + p_y b - p_z$

$h^* = (a, b, c)$, so the inequality, $c > ap_x + bp_y - p_z$, then states that the point h^* has larger z coordinates than the point of the plane p^* below it at (a, b) . Hence h^* lies above p^* .

Therefore, p lies above $h \leftrightarrow h^*$ lies above p^*

11.10C

$UH(P)$ is the set of upper faces of $CH(P)$

$LE(P^*)$ is the lower envelope of the dual planes

p^* appears on $LE(P^*)$ meaning that the plane p^* contributes a 2 dimensional face to the lower envelope.

Need to prove that p is a vertex of $UH(P) \leftrightarrow p^*$ appears on $LE(P^*)$

First prove if p is a vertex of $UH(P)$ then p^* appears on $LE(P^*)$

Since p is vertex of $UH(p)$, then there is family of non vertical supporting planes $h-p$ such that every other point of P lies on or below h .

For each plane h

$p \in h$, $h^* \in p^*$ by incidence

Every $q \in P \setminus \{p\}$ lies below h , h^* lies below q^* by order preservation.

Every dual point h^* lies on the lower envelope and also lies on the plane p^* .

Therefore collection of dual points forms 2D patch on p^* that belongs to $LE(P^*)$.

Hence, p^* appears on $LE(P^*)$

Now prove if p^* appears on $LE(P^*)$ then p is a vertex of $UH(P)$

If p^* appears on $LE(P^*)$, then there is a 2D patch $R \subset p^*$ such that every point $h^* \in R$ lies on lower envelope.

Pick any of the h^* in the interior of R , and let $h = (h^*)^*$

Since $h^* \in p^*$, the incidence property then states $p \in h$.

Because h^* lies on lower envelope, it lies below every other dual plane q^* .

Through order preservation property, this means every other point $q \in P \setminus \{p\}$ lies on or below h

So h is a supporting plane of $CH(P)$ through p . Since whole 2D patch of such dual points h^* , there is a 2 parameter family of supporting planes through p . This happens when p is a vertex of the upper hull. If P is in interior of upper edge or facet, then the supporting planes through p would form a 1D family or a single plane.

Therefore, p is a vertex of $UH(P)$ and the proof is proven correct.

11.10D

$p = (p_x, p_y, p_z)$ and $q = (q_x, q_y, q_z)$

Dual planes are $p^* : z = p_x x + p_y y - p_z$ and $q^* : z = q_x x + q_y y - q_z$

Need to prove that pq is an edge of $UH(P) \leftrightarrow p^*$ and q^* share edge on $LE(P^*)$

First prove that if pq is an edge of $UH(P)$ then p^* and q^* share edge on $LE(P^*)$

If pq is an edge of $UH(P)$ then there is a family of supporting planes h for the upper hull that contain the line through p and q , and all points of P lie on or below each plane.

Since $p, q \in h$, incidence property is $h^* \in p^* \cap q^*$

Since all $r \in P$ lies on or below h , order preservation property states h^* lies on or below every r^* .

So h^* lies on the lower envelope of $LE(P^*)$ and therefore p^* and q^* have their own pieces of the lower envelope and share an edge on $LE(P^*)$

Now need to prove that if p^* and q^* share an edge on $LE(P^*)$ then pq is an edge of $UH(P)$

p^* and q^* share an edge e on $LE(P^*)$

Choose a point h^* in the relative interior of edge e , and let $h = (h^*)^*$.

Since $h^* \in p^* \cap q^*$, the incidence property states that $p, q \in h$

Since h^* lies on the lower envelope, it lies on or below every dual plane r^* , so through order preservation all points $r \in P$ lie on or below h

Therefore h is a supporting plane of the upper convex hull through p and q . There is a 1D set of dual points h^* along e , there is a 1 parameter family of supporting planes through p and q . Therefore, the exposed face determined by p and q is 1D, so pq is an edge of $UH(P)$

So, it is proven that pq is an edge of $UH(P) \leftrightarrow p^*$ and q^* share edge on $LE(P^*)$

11.10E

Need to show that $p_1, p_2, \dots, p_k$ are the vertices of a facet f of $UH(P)$ if and only if $p_1^*, p_2^*, \dots, p_k^*$ supports the facets of $LE(P^*)$ that share a common vertex

Forward Direction:

Assume $p_1, \dots, p_k$ are the vertices of a facet f of $UH(P)$

Then there is a unique supporting plane h that contains that facet, so $p_i \in h$ for all $i = 1, \dots, k$ and every other point of P lies on or below h

Through incidence, $h^* \in p_i^*$ for all $i = 1, \dots, k$

Through order preservation and how every point of P lies on or below h , the point h^* lies on or below every dual plane in P^* . Hence h^* is a vertex of the lower envelope $LE(P^*)$

Near the vertex h^* , each plane p_i^* contributes one facet of the lower envelope, and all these facets meet at the common vertex h^* .

Therefore, $p_1^*, \dots, p_k^*$ supports facets of $LE(P^*)$ that share a common vertex.

Reverse Direction Proof:

$p_1^*, \dots, p_k^*$ supports facets of $LE(P^*)$ that share a common vertex v .

Let $h = v^*$

Since $v \in p_i^*$ for every i , through incidence property, $p_i \in h$ for all $i = 1, \dots, k$;

Since v is vertex of lower envelope, it lies on or below every plane in P^* .

Through order preservation, every point of P lies on or below the plane h .

So h is a supporting plane of $CH(P)$ and the set $CH(P) \cap h$ is a face of the upper convex hull.

Since a lot of lower-envelope facets meet at this v , the face is 2D, so it is a facet.

Vertices are points $p_1, \dots, p_k$.

Therefore, $p_1, p_2, \dots, p_k$ are the vertices of a facet of $UH(P)$ iff $p_1^*, p_2^*, \dots, p_k^*$ support facets of $LE(P^*)$ sharing a common vertex